Investigating the Limits of the Travelling Fire Methodology

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Investigating the Limits of the Travelling Fire Methodology - Progress Report

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Abstract

The selection of design fires for structural fire engineering has faced scrutiny in recent times due to the trend towards larger compartments. The design of larger compartments questions the applicability of the traditional structural fire engineering methods and the concept of travelling fires has been proposed as an alternative.

This project seeks to investigate the input parameters of the existing travelling fire methodologies and identify their limitations to further understand the appropriateness of this method.

This progress report provides the status of the project as well as the proposed work to be undertaken over the remainder of the project. The background to the project and the research aims of the project are first outlined to provide the context of the project. A literature review focusing on the current state of compartment fire research has been conducted and reviewed the traditional compartment fire framework and the more recent travelling fire concepts to provide the context of the current research state.

An overview of the proposed research process and methodology has been provided with spreadsheet models implementing the MQH correlation and TFM to be adopted to study the impact of input parameter variation. Preliminary results from the spreadsheets showed similarities with other sources and identified the potential shortcoming in the proposed methodology which shall be extended over the rest of the project.

A risk assessment for the remainder of the project has found it to be low risk with data analysis and project management identified as critical areas. An indicative project plan to assist the planning and coordination of the remainder of the project has also been developed to address the project management risks.

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1. Introduction

1.1. Project Background

Structural fire engineering involves the of design elements to achieve an adequate level of fire resistance and requires the selection of design fires on which to base the exposure (Buchanan & Abu, 2017). However, recent architectural trends have moved towards larger building developments which have questioned the limits of traditional structural fire engineering methods.

Traditional methods have been based on the compartment fire framework which considers a fully developed fire and hence assumes a uniform burning of fuel and homogenous temperatures. The trend towards larger open plan compartments challenges this inherent assumption of uniform temperature and thus the phenomenon of travelling fires has been proposed as an alternative to the traditional methods.

The concept of travelling fires involves the development and burning of a fire within a localised area that eventually spreads to the adjacent fuel and continues to burn after the fire in the initial location has burnt out. This results in a fire burning over a localised area which “travels” across a fuel path. This phenomenon may apply in either horizontal or vertical configurations, however this research focuses on horizontal travelling fires across the floor plate of a compartment.

The concept of travelling fires has been extensively researched as evident by the available literature and publications on the topic. Multiple methodologies and models have been proposed and developed seeking to characterise the behaviour of travelling fires. Despite the extensive research, it has been identified that the strict limitations of the existing travelling fire methodologies are not clearly articulated.

1.2. Research Aims & Objectives

The aim of this project is to investigate the input parameters of the current travelling fire methodology and improved variations and identify the limitations of these methodologies.

This research aim shall be achieved through undertaking the following objectives:

1. To review the conditions required for a travelling fire to occur within a compartment.
2. To review the underlying assumptions of the travelling fire methodologies and understand their impact on the outputs.
3. To review the input parameters of the travelling fire methodologies and understand how they influence the potential for a travelling fire.
4. To study the impact of the identified input parameters and derive limits based on physical limitations or experimental observations documented in literature.

It is emphasized that this project does not seek to address the identified limitations, but rather further the understanding of the existing methodologies.

2. Literature Review

2.1. Compartment Fire Dynamics

2.1.1. Fuel Loads

The fuel load within a compartment has significant implications on fire behaviour irrespective of the different methodologies used to describe the fire dynamics as it governs the amount of heat that can be potentially generated (Khan et al., 2021).

Current traditional and travelling fire approaches both assume an even fuel load distribution across the floor plate, with the main difference being that in traditional methods, all fuel considered to ignite and burn simultaneously whilst in travelling fires the fuel is burnt over a staggered duration as the fire spreads.

2.1.2. Ventilation

Another major component of fire behaviour is the oxygen available to partake in the combustion reaction. Kawagoe establish the concept of compartment fires which linked the compartment temperatures with ventilation. This was further researched by Thomas et al and also by Harmathy who both classified two regimes:

- Ventilation-controlled (or Regime I) where the oxygen available to participate in the reaction is the limiting factor for the combustion reaction.
- Fuel-controlled (or Regime II) where there is sufficient oxygen such that the combustion reaction is limited by the fuel available to participate in the reaction.

The distinction between these regimes plays a crucial role in the fire dynamics within a compartment. The ventilation area and in particular the opening factor is noted to relate to the evolution of the gas phase temperature as combustion product may be expelled via the opening (Torero et al., 2014), thus impacting the fire development and spread rate.

2.1.3. Spread Rate

The ratio between flame spread (V_S) and burnout rate (V_{BO}) are noted to define the fire dynamics within a large compartment (Gupta et al., 2021) which can be identified as three distinct modes:

1. $\frac{V_S}{V_{BO}} \rightarrow \infty$ which represents fully developed fires where flashover has occurred.
2. $\frac{V_S}{V_{BO}} > 1$ which represents growing fires where flashover has not occurred.
3. $\frac{V_S}{V_{BO}} \approx 1$ which represents travelling fires.

The modes of fire spread noted above are not independent and transition between the modes is possible. In all cases a fire is expected to start as a mode 2 growing fire before

transitioning to either mode 1 or 3. In the case of transition to mode 1, a fully developed fire results due to the increased radiation to unburnt fuel leading to an increased spread rate which is much greater than the burnout rate. Conversely mode 3 results when the spread rate and burnout rate are approximately equal leading to a travelling fire being established.

2.1.4. Flashover

Traditional structural fire engineering assumes a flashover has occurred which is defined as *"the rapid transition to a state of total surface involvement in a fire of combustible material within an enclosure"* and is used to indicate the transition from the growth phase to fully developed stage (Karlsson & Quintiere, 1999). Another description of flashover is as a thermal instability (Drysdale, 2011).

The conditions to indicate the onset of flashover are poorly defined which has been identified as surprising given its importance (Bishop & Drysdale, 1998). Despite the unclear definition, some common criteria for flashover have been identified (Karlsson & Quintiere, 1999) and broadly consistent with multiple experimental observations (Babrauskas et al., 2003):

- Compartment gas temperatures exceed 500-600°C.
- Radiant heat flux over 20 $kW.m^{-2}$ to the floor.
- External flaming through the compartment openings is observed.

As flashover is also considered the clearest indicator of untenable conditions in a compartment (Peacock et al., 1999) there has been significant research into predicting when flashover may occur. McCaffrey, Quintiere and Harkleroad developed a correlation based on experimental data that describes the relation between the compartment geometry, boundaries and the heat release rate to achieve a gas temperature of 500°C which is used as the criteria for flashover (McCaffrey et al., 1981).

It is considered that the common indicators for flashover as identified above would be inadequate in describing the conditions in large compartments as fires would be expected to be localised to certain areas within the compartment. This is further reinforced by the recognition that post-flashover fires would be unlikely to occupy the whole space at once (Buchanan & Abu, 2017). These findings question the design process for large compartments if the traditional approaches which assume whole compartment involvement are not strictly applicable.

2.1.5. Categories of Fire Models

The continual research into fire science and engineering has led to the development of multiple models to understand and predict the fire dynamics in a compartment. These models have been categorised by based on their applicability (Khan et al., 2021). This distinction between uniform temperature (category I) and non-uniform temperature (category II) models highlights the variability in compartment fires and the extensive research undertaken. These categorise are summarised in Table 2.1.

Table 2.1: Summary of Categories of Fire Models

Category	Description & Characteristics	Examples of Models
I	Uniform temperatures	Standard temperature-time curves Parametric fire curves
II	Non-uniform temperatures	Localised burning Travelling fires
III	Realistic temperature distributions	Experimental data CFD Models

As this research is aimed at identifying the limitations of the travelling fire methodologies which fit into category II, the focus shall be placed on this category, however a brief review of the traditional approaches which are classified under category I will also be undertaken.

2.2. Traditional Approaches

The traditional compartment fire framework is based on an energy balance within a compartment during the growth and fully developed phases. The traditional models currently adopted in structural fire engineering for determining fire resistance of elements involve two main types of temperature-time curves which are also based on experiments conducted using relatively small scale rectangular compartments (Jonsdottir et al., 2010):

- Standard Temperature-Time Curves
- Parametric Fire Curves

These methods assume a uniform temperature within the compartment which is justified by the expected small temperature gradients in vertical and horizontal direction (Horová et al., 2013). The inherent assumption of these methods is that the structural fire resistance is most tested under a fully developed fire whereby the whole compartment is burning (Gernay, 2023).

2.2.1. Standard Temperature-Time Curve

The standard temperature-time curve provides a globally accepted fire resistance test criteria that within multiple standards such as AS 1530.4 and ISO 834. The curve provides the standard heating regime that a furnace test is required to expose the test member to. The standard temperature-time curve is solely a relation between the time and temperature and does not account for any compartment characteristics.

Furthermore, the standard temperature-time curve has been questioned due to the unrealistic nature of the curve compared to that of a real fire and the potential limits associated with this (Gales et al., 2021). It is also noted that there are multiple other curves which describe certain fuel loads or parameters, however for the purposes of this project any comparisons will be made with the standard ISO 834 temperature-time curve.

2.2.2. Eurocode Parametric Fire Curves

The parametric temperature-time curve is defined in Eurocode 1 and seeks to provide an alternative to the standard temperature-time curve and accounts for the compartment geometry as well as the decay phase of a fire. Furthermore, the parametric fire is noted to be applicable to compartments with floor areas up to 500 m^2 and 4 m in height thus highlighting that they are not applicable to large compartments.

The parametric temperature-time curve is defined by different formulae for the heating and cooling phases as detailed in Eurocode 1 BS EN 1991-1-2. Comparing the parametric fire curve with the standard temperature-time curve, the key difference is the consideration of the decay phase considered in the parametric curve as shown in Figure 2.1. Furthermore, the parametric fire curve is specific to a compartment the compartment geometry and openings are inputs into the curve.

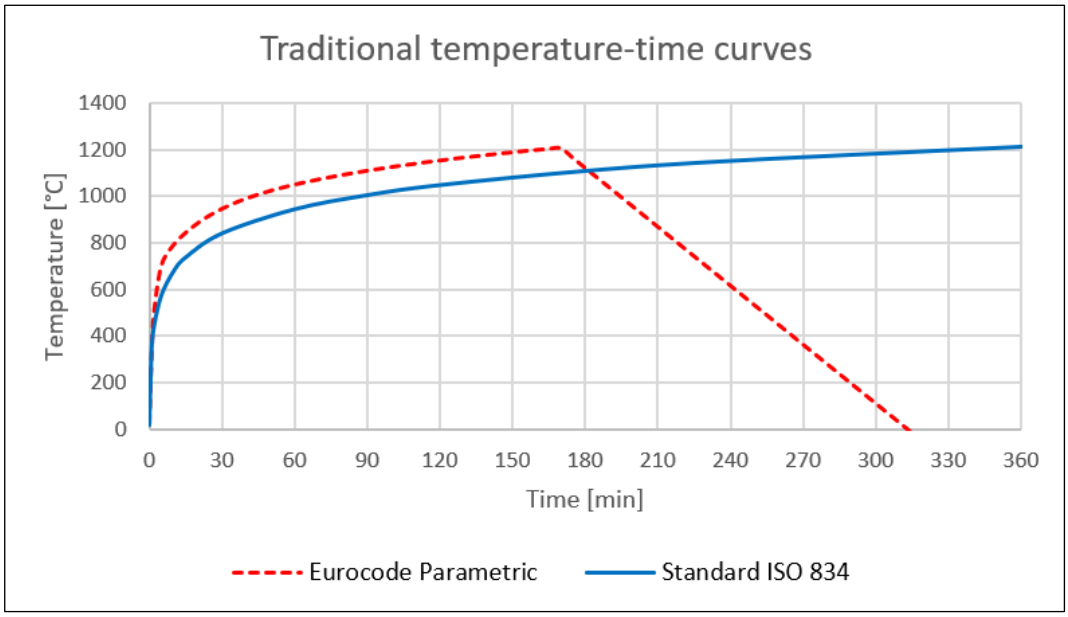


Figure 2.1: Examples of traditional temperature-time curves adopted in structural fire engineering

2.3. Travelling fires

Unlike traditional approaches which assumes uniform temperature across the compartment over the burning period once fully developed, the travelling fire methodologies account for non-uniform temperatures based on the spatial and temporal evolution of the fire.

2.3.1. Conditions for a Travelling Fire

2.3.1.1. Ventilation Requirements

Travelling fire models have been developed which assume either ventilation-controlled or fuel-controlled conditions (Dai et al., 2017). Ventilation conditions can also impact the travelling behaviour due to the smoke accumulation (Chu et al., 2023).

The different models developed which assume either regime fire indicate that it cannot currently be conclusively stated what ventilation conditions are required for a travelling fire to occur. This is similar to traditional methods such as the standard temperature-time curve assuming a fully developed fire where ventilation conditions are not always considered.

2.3.1.2. Fuel Load, Distribution & Burning Rates

In order for a travelling fire to develop it is required that fuel be available for the fire to spread across and thus similar to traditional compartment fire approaches an even and constant fuel distribution is assumed. Whilst accepted to not necessarily reflect actual structures or compartments, it is a nonetheless a critical assumption.

The type of fuel load is also a critical for current travelling methods in order to have a flame spread rate to burnout rate ratio of 1 which is representative of a travelling fire. Where this criteria is not met, two scenarios are noted; a travelling fire does not develop, or a travelling fire establishes which continues to grow and transition into a fully-developed fire.

2.3.2. Types of Travelling Fires Models/Methodologies

There are multiple travelling fire methodologies and compartment fire frameworks that have been developed. Some of the models are noted to have been developed independently of each other whilst others are noted to be improvements over the previous model. However, the key objective of these models is to provide a methodology for which to assist the design phase of these large compartments.

The travelling fire methodologies do not typically yield a single output for design considerations but rather a set of curves which are then often then used as input parameters for heat transfer and structural analysis. However despite the extensive research, it has been identified travelling fires do not have the same amount of experimental validation that is typically required (Rush et al., 2021).

2.3.2.1. Large Fire Cell Method (LFM) - Clifton Model

An early model that sought to describe a travelling fire was the Large Fire Cell Method (LFM) developed by Clifton as part of the HERA programme in New Zealand (Clifton, 1996). This model divides the multiple areas each subject to individual temperature-time curves.

Clifton acknowledged that the LFM should mostly function as a research tool due to the required assumptions. This admission combined with the limited evidence of further development or application of the model (Stern-Gottfried & Rein, 2012a) has led to the LFM method not being further investigated as part of this research.

2.3.2.2. Travelling Fire Methodology (TFM)

The Travelling Fire Methodology (TFM) developed by Rein et al (2006) and later refined by Stern-Gottfried and Rein (2012) was developed as a methodology independently of the Cliton model to assist structural design.

The TFM is based on the spatial discretisation of a compartment into bays whereby the temperature fields may be determined based on the assumed fire size, fuel loads and burning times. The determination of these temperatures utilises various assumptions and analytical correlations.

The thermal fields are divided into two regions which are identified based on the relative position with respect to the fire location. These regions are referred to as the near-field and far-field. The near-field is associated with the burning region where direct exposure to the flames is expected. The far-field is defined as the remote locations where heating is due to exposure to the smoke layer and not flames (Stern-Gottfried & Rein, 2012a). In particular the far-field temperature utilises Alpert's correlation to determine the temperature fields. Although Alpert's correlation is also applicable to the near-field regions, the TFM assumes a constant near-field temperature of 1200°C (Stern-Gottfried & Rein, 2012b).

Another feature of the TFM is the constant burning and spread rate which is in line with the criteria of a spread rate to burnout rate ratio of 1. However, the TFM does not consider any transitional states where the fire growth may be accelerating and approaching a fully-developed phase. Furthermore, the TFM is defined for a predetermined fire path from one end of the compartment to another, and thus inherently assumes the fire starts at one end.

When compared to the traditional methods, the TFM differs in a number of ways. The first is the input parameters which considers the compartment geometry but does not account for the ventilation. Another major difference is the outputs where unlike traditional methods the TFM also provides a set of fire curves that covers a range of possible fires within the compartment (Jonsdottir et al., 2010)

Despite the TFM providing a methodology to describe the set of temperature profiles within a compartment, it is criticized for not conserving mass or energy. Although this criticism is valid, traditional methods particularly the standard temperature-time curve has also been criticised for the same omission.

2.3.2.3. Improved Travelling Fire Methodology (iTFM)

The improved travelling fire methodology (iTFM) was developed by Rackauskaite (2015) and is based on the TFM by Stern-Gottfried and Rein. The key improvement of the iTFM over the original TFM is the consideration for 'flame flapping' which changes the near-field temperature assumed in the model. The idea of the flame flapping is to account for the dynamic behaviour of a fire which would vary the flame location and thus impact caused by the flame. This phenomenon is accounted for by calculating the flame angle and adjusting the assumed near-field temperature to between 800-1200°C instead of assuming a fixed 1200°C (Rackauskaite et al., 2015).

Similar to the TFM, a fixed burning rate and duration is assumed in the iTFM which also makes no improvement to the lack of conservation of mass and energy. As a result, the input parameters between the TFM and iTFM are the same there have been no identified differences between the applicability of either model over the other for a given compartment.

2.3.2.4. Flame extension Travelling Fire Methodology (fTFM)

Another improved variation to the TFM developed by Stern-Gottfried and Rein is the flame extension travelling fire methodology (fTFM) (Heidari et al., 2020). Similar to the iTFM, the fTFM seeks to improve upon the assumptions regarding the near-field temperatures. When the flames directly interact and impinge on the ceiling of the compartment it noted to change the ceiling jet behaviour and impact the temperature fields at this turning region (Zeng et al., 2023). The fTFM considers this flame extension phenomenon by adjusting the calculation of the near-field temperature.

This interaction results in part of the flame deflecting horizontally resulting in an extended near-field region since the flames are now directly impacting a larger region (Heidari et al., 2020). Compared to the iTFM, the fTFM does not dispute any of the previous assumptions or simplifications and instead adds the consideration of flame extension to the model.

Another improvement the fTFM made compared to the TFM and iTFM is the translation from a temperature-time curve to produce a heat flux curve which is more directly applicable for heat transfer calculations and structural analysis (Heidari et al., 2020).

However, similar to the previous TFM and iTFM, the lack of conservation of mass and energy is still maintained. These consistent shortcomings and the same underlying assumptions suggest that there are few differences between the applicability of one method from another.

2.3.2.5. Extended Travelling Fire Method (ETFM) Framework

Another design methodology that has been developed independently of the Clifton model or Rein model is the Extended Travelling Fire Method (ETFM) framework. The ETFM is based on Hasemi's localised burning model which is assumed to travel a predetermined path combined with zone models (Dai et al., 2017).

The ETFM framework considers both energy and mass conservation which differs and addresses some of the concerns of the previously discussed TFM models. The inherent differences and implementation of the ETFM framework compared to other methods, makes comparison between them difficult and thus the ETFM shall not be investigated further as part of this research.

2.3.3. Summary of Methodologies

Whilst a summary of the various travelling fires has been presented in Table 2.2, this project shall focus on the application of the TFM and improved variations (i.e. iTFM and fTFM).

Table 2.2: Summary of Travelling Fire Methodologies

Fire Model	Characteristics & Assumptions	Reference
LFM Clifton Model	Temperature-time curve for whole fire cells. Fire is ventilation-controlled. Mass and energy conservation not considered.	(Clifton, 1996)
TFM	Single constant near-field temperature of 1200°C. Far-field temperature determined by Alpert's correlation. Constant burning rate along a predefined path. Fire is fuel-controlled. Mass and energy conservation not considered.	(Stern-Gottfried & Rein, 2012b)
iTFM	Flame flapping considered giving near-field temperature of 800°C - 1200°C. Far-field temperature determined by Alpert's correlation. Constant burning rate along a predefined path. Fire is fuel-controlled. Mass and energy conservation not considered.	(Rackauskaite et al., 2015)
fTFM	Translates temperature-time curves to heat flux curves. Flame extension caused by flame interactions with ceiling considered. Far-field temperature determined by Alpert's correlation. Constant burning rate along a predefined path. Fire is fuel-controlled. Mass and energy conservation not considered.	(Heidari et al., 2020)
ETFM	Hasemi's local fire model implemented for near-field temperature. Far-field temperature determined by two-zone model. Mass and energy conservation considered.	(Dai et al., 2020)

3. Proposed Research Methodology

3.1. General

An overview of the methodology adopted to the investigate the limitations is illustrated in Figure 3.1.

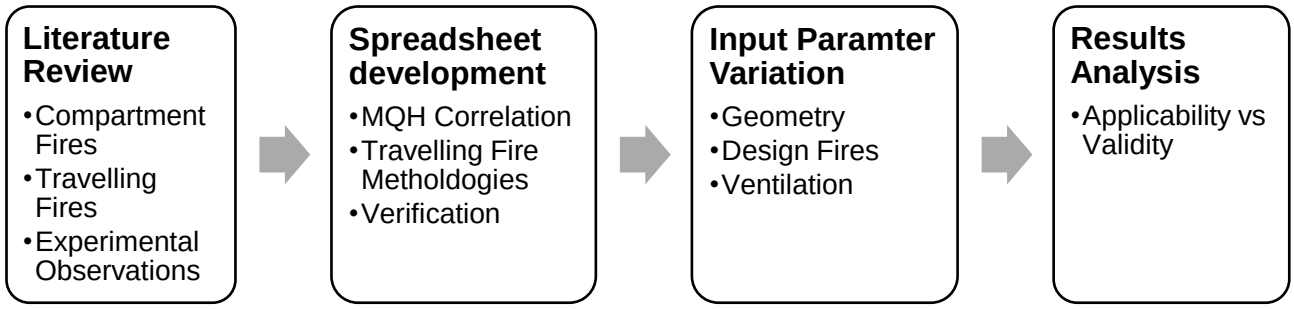


Figure 3.1: Overview of the Research Process

3.2. Literature Review

The literature review documented in Section 2 focused on the general concept of compartment fires in addition to the concept of travelling fires and the different travelling fire methodologies currently developed. This review has provided a foundation for the remainder of the project.

As part of the research on-going literature review is expected to be required. This additional literature review shall have a continual focus on both traditional compartment fires and travelling fires, and in determining how the assumptions of each impact the limitations of the methodologies and how they may also correlate with experimental observations.

3.3. Spreadsheet Development

To study the input parameters of the TFM, spreadsheet models adopting analytical relations described in the MQH flashover correlation and TFM variations shall be developed.

Spreadsheet models have been proposed for this project due to the ability to rapidly test multiple configurations and input parameters. Furthermore, the design methodologies to be investigated are noted to be based on analytical solutions which utilise relatively simple calculations suitable for implementation within a spreadsheet.

Other methods of investigation including the use of CFD modelling through FDS have been considered for the study, however the inherent simplifications made for the TFM (and other variations), and the increased computational cost associated with CFD modelling have deemed it unnecessary compared to the potential to test multiple configurations using developed spreadsheets implementing analytically derived formulae.

3.3.1. McCaffrey, Quintiere & Harkleroad Correlation

The McCaffrey, Quintiere & Harkleroad correlation to estimate the time required to flashover shall be utilised. The intent of the MQH model will be to assist in determining the limits of applicability for the TFM as the inherent nature of a travelling fire implies that flashover will not occur. In this project, it is proposed to use the MQH model to determine if flashover is possible for given compartment parameters and an assumed heat release rate of a fire to be also adopted in the TFM.

The MQH correlation considers the compartment geometry and opening areas and is defined as follows:

$$Q_{flash} = 610\left(h_k A_w A_o \sqrt{H_o}\right)^{1/2}$$

Where

Q_{flash} heat release rate at flashover [kW]

A_w total surface area of solid boundaries including door openings [m^2]

A_o opening area [m^2]

H_o height of compartment [m]

h_k effective heat transfer coefficient of the solid boundaries [$kW.m^{-2}.K^{-1}$] where

$$h_k = \sqrt{\frac{k\rho c}{t}}$$

$k\rho c$ thermal inertia of compartment boundaries [$kW^2.s.m^{-4}.K^{-2}$]

t time [s]

The MQH correlation enables an estimation of the heat release rate required to achieve temperatures indicating flashover in a specified time frame. Depending on whether the heat release rate of the fire is greater than the heat release rate required for flashover and the local burning time four scenarios have been identified:

1. $Q > Q_{flash}$ and $t_{BO} < t_{FO} < t_{total}$ which represents a growing fire which develops into a travelling fire before transitioning into flashover.
2. $Q > Q_{flash}$ and $t_{BO} < t_{total} < t_{FO}$ which indicates no flashover.
3. $Q > Q_{flash}$ and $t_{FO} < t_{BO}$ which indicates a growing fire and flashover occurring before a trailing edge establishes.
4. $Q < Q_{flash}$ which indicates flashover cannot occur.

Based on the cases identified above, a travelling fire would occur in either case 1, 2 and 4 whilst case 3 could be considered to represent a fire as described by the traditional compartment fire framework. However, the TFM models which are to be the focus in this work would require a case 2 or 4 fire as the TFM does not account for an accelerating spread rate that would transition into flashover.

3.3.2. TFM and Variations

The TFM, iTFM and fTFM shall be implemented after it has been determined if flashover is possible using the MQH correlation for a given set of compartment parameters. The outputs of these travelling fire methodologies shall then be compared with previous experimental observations.

As previously noted, the outputs of the TFM are temperature-time curves specific to a point within the compartment. This research shall focus on the temperature-time curves established for three points representing the central point and two ends of the compartment.

The iTFM and fTFM shall also be implemented as these improvements adjust some of the underlying assumptions particularly around the near-field temperature. These alterations thus potentially influence some the limitations to be identified.

3.3.3. Verification

The verification of any models with selected tools is critical to determine the reliability of any conclusions. In this project the developed spreadsheets shall be verified through literature reviews. As the proposed spreadsheets shall be implementations of existing methodologies, there are documented results of their application in various literature sources which shall form the basis for the verification. The developed models for this project shall recreate the results found in these sources to confirm the spreadsheets are configured appropriately.

3.4. Input Parameter Variation

The aim of this research is to determine the limits of the existing travelling fire methodology and improved variations which requires the input parameters to the model to be investigated and parameterised such that the impact of each variable is known.

The MQH model as previously discussed shall be used initially to determine if flashover is possible for a given set of input parameters. Once determined if flashover is possible, the TFM and subsequent variations shall be implemented for the same parameters.

3.4.1. Geometric Parameters

The compartment geometry is an essential parameter when determining the fire behaviour within it. In the case of the TFM, the compartments are assumed to rectangular prisms in nature with constant lengths, widths and heights.

3.4.1.1. Floor Area

It is widely accepted that the concept of travelling fires would be applicable to large compartments, however it has been identified that the criteria of when a compartment is considered ‘large’ is not well defined (Stern-Gottfried & Rein, 2012a). This project seeks to further investigate this observation and determine the geometric limits of applicability of the TFM which does not appear to have any current specified compartment size limitations.

3.4.1.2. Geometric Aspect Ratio

In addition to investigating different overall compartment floor areas, the aspect ratio between the length and width of a compartment shall also be investigated. There is limited research into the impact of the compartment aspect ratio on a travelling fire and as such its importance is currently unknown.

For this research the aspect ratio shall always be expressed as a value greater than 1 to reflect the TFM's assumption that the entire width is burning. It is proposed to investigate aspect ratios ranging from 1 (equal length and width) to 5 (length five times the width).

3.4.1.3. Height

The height of a compartment is also noted to have a significant influence on the fire dynamics due to the impact on plume behaviour. This importance in the TFM is reflected by the utilisation of Alpert's correlation to determine the far-field temperatures.

$$T_{max} - T_{\infty} = 5.38\left(\frac{(\dot{Q}/r)^{2/3}}{H}\right)$$

Where

T_{max} Temperature of ceiling gas [°C]

T_{∞} ambient temperature [°C]

$\dot{Q}$ heat release rate [kW]

r horizontal distance to the fire centre line [m]

H height of the compartment [m]

Furthermore, the ceiling height may influence the potential for direct exposure of the ceiling to flames and thus change the near-field temperature as identified in the iTFM and fTFM.

3.4.1.4. Opening Sizes

The TFM is noted to not account for the opening areas in a compartment with the inputs based on compartment size and fire parameters. This lack of ventilation consideration creates currently potential unknown limitations with respect to the applicability of the TFM as the possibility of flashover is not inherently considered.

To investigate the impact of openings, it is proposed to adopt a base opening size which shall be implemented for all test cases. Additional openings shall be assumed within the model to increase the opening factor within the specified range of 0.02-0.20 in line with the limits of the Eurocode parametric fire curve.

3.4.2. Fire Parameters

The key parameters in relation to the fire in the TFM is the overall area of the fire, fuel load density and the heat release rate.

3.4.2.1. Burning Area

The TFM assumes a constant percentage of the floor area is burning at any time. This aligns with the travelling fire criteria for a spread rate vs burnout rate ratio of 1, however in the TFM the percentage of the floor area burning is defined as an input parameter. It has been identified that a structure experiences the most challenging conditions from a travelling fire burning over approximately 25% of the floor area (Law et al., 2011), however other sources have stated 5-20% (Stern-Gottfried & Rein, 2012b). A burning area range of 5-30% shall be investigated in line with the most challenging conditions.

3.4.2.2. Fuel Load Density

The TFM and variations assumes a fixed fuel load density similar to the traditional parametric fire curve. Fuel load densities of 285-1,500 $MJ.m^{-2}$ representative of sparsely furnished areas to densely loaded spaces were tested in the TFM development (Stern-Gottfried & Rein, 2012b). This same range shall be adopted in this project.

3.4.2.3. Heat Release Rate

The TFM and variations requires a fixed heat release rate per unit area to be predefined, and since the models also assumes a constant burning area the total heat release rate of the fire would also be constant. The MQH correlation is noted to also represent a fixed heat release rate, which enables the fire heat release rate to be compared with the flashover heat release rate.

The typical heat release rates of 200-800 $kW.m^{-2}$ representative of non-industrial buildings tested in the TFM development (Stern-Gottfried & Rein, 2012b) shall also be adopted during the investigation.

The constant fuel load density and heat release rate per unit area results in the local burning time also being calculated as a constant which aligns with the travelling fire criteria for a spread rate vs burnout rate ratio of 1 and constant burning area.

3.4.3. Summary of Parameters

It is currently proposed to investigate different parameters independently with other variables held at constant base values. The range of parameter values to be investigated are summarised in Table 3.1

Table 3.1: Summary of Input Variables

Parameter	Parameter Type	Proposed Range to Investigate	Comment
Compartment Floor Area	Geometric	50-2,000 m^2	50 m^2 is considered representative of small compartments. 2,000 m^2 is considered representative of any “large” compartment.
Length-to-width Aspect Ratio	Geometric	1-5	Square compartment with equal length and width. Rectangular compartment with 5 times the length to width.
Compartment Height	Geometric	3-18 m	3 m is considered representative of standard building level heights. 18 m is the maximum height Alpert’s correlation is derived for (Zeng et al., 2023).
Ventilation Opening	Geometric	0.02-0.20 $m^{0.5}$	Valid range for traditional parametric fire curve.
Burning Area	Fire	5-30%	Range for most challenging conditions to structure.
Fuel Load Density	Fire	285-1,500 $MJ.m^{-2}$	Range of fuel load density in non-industrial buildings tested by Stern-Gottfried for the TFM
Heat Release Rate per unit area	Fire	200-800 $kW.m^{-2}$	Range of HRRPUA in non-industrial buildings tested by Stern-Gottfried for the TFM
Thermal inertia of compartment boundary	Material	1,160 $J.m^{-2}.s^{-0.5}.K^{-1}$	Value typical of concrete which shall be fixed for the project

3.5. Results Analysis

The concept of limitations for models shall be explored from two aspects. The first aspect shall investigate the inputs and focus on where models may no longer be applicable for given input parameters. The second aspect focuses on the outputs and where models may no longer produce results which are realistic or valid of the situation. These two aspects shall be considered as the applicable limitations for inputs and the validity limitations of outputs.

3.5.1. Applicable Limitations

Applicable limits in this research shall refer to the model input parameter limitations. The TFM is not currently understood to have any specified limitations or inherent checks of the input parameters. Conversely the Eurocode parametric fire has specific defined upper limits such as being applicable for compartments up to 500 m^2.

To address this, the MQH correlation shall be adopted to provide an indication of the potential for flashover for any given parameters. In instances where the flashover is found to be expected, then it can be concluded that the TFM is not applicable. However, this shall not determine the validity of the TFM in instances where flashover would not occur and a travelling fire is expected to develop.

3.5.2. Validity Limitations

Validity limitations shall refer to the output parameters investigate the scenarios where results may be unrealistic or invalid for the given input parameters. In instances where the assumed parameters indicate the potential for a travelling fire, then the output values of the TFM variations shall be further investigated.

Unlike the applicable limitations, validity limitations shall not be determined based on the potential for an event (i.e. flashover), but rather the conditions and behaviour of the fire compared to physical conditions identified from experimental observations or practical applications for given structures.

4. Preliminary Results

4.1. Preliminary Spreadsheets

At this stage two preliminary spreadsheet models have been setup which are the MQH correlation and the TFM by Stern-Gottfried. These spreadsheets shall be further refined and verified before any further testing of input parameters.

Further spreadsheets implementing the iTFM and fTFM shall be developed discuss the potential differences in application between them and the original TFM by Rein.

4.1.1. MQH Correlation

The MQH correlation spreadsheet the documented results recreated to check that the input parameters have been understood. The initial results are shown in Figure 4.1 and show a reasonable level of similarity between them indicating that the model has been implemented as designed.

Furthermore, a sample MQH heat release rate-time curve using the preliminary spreadsheet shown in Figure 4.2 depicts the regions indicating whether flashover would be expected.

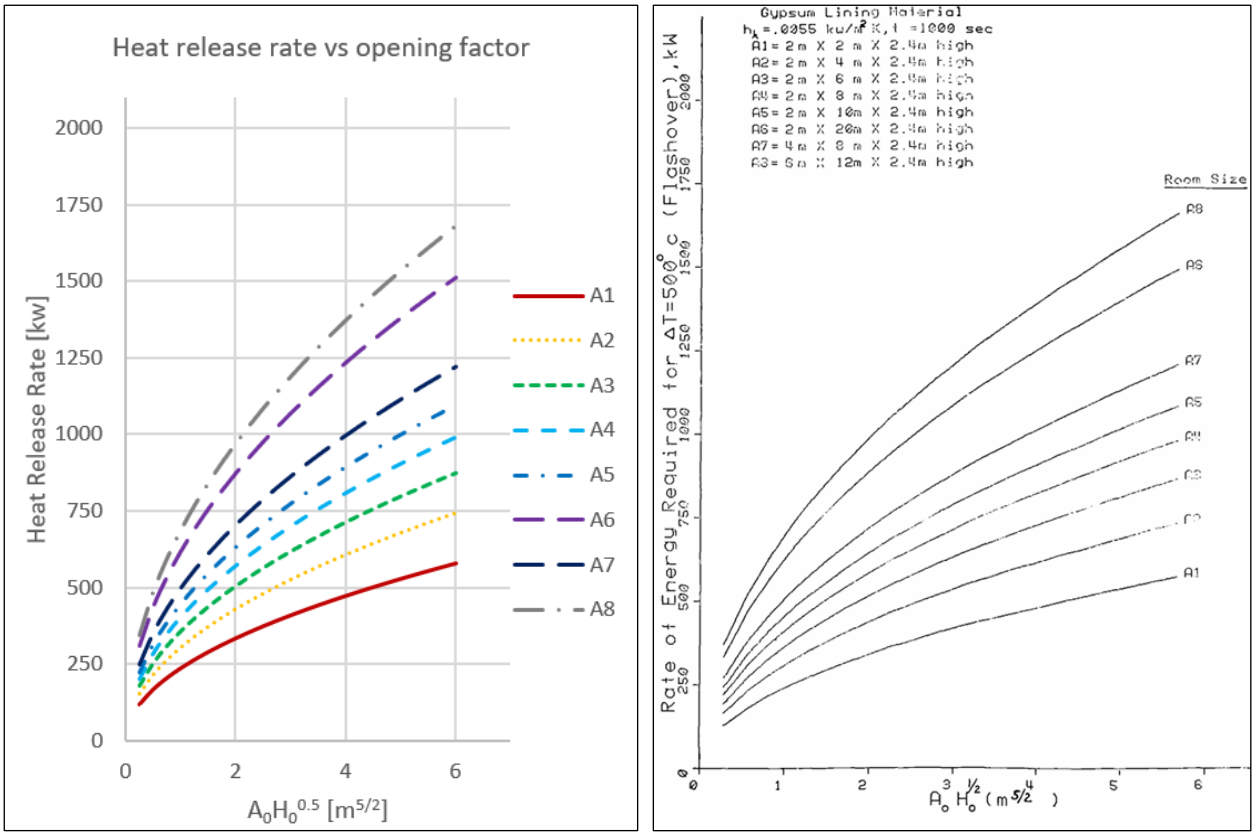


Figure 4.1: Preliminary MQH Results: Developed spreadsheet (left) and (McCaffrey et al., 1981)(right)

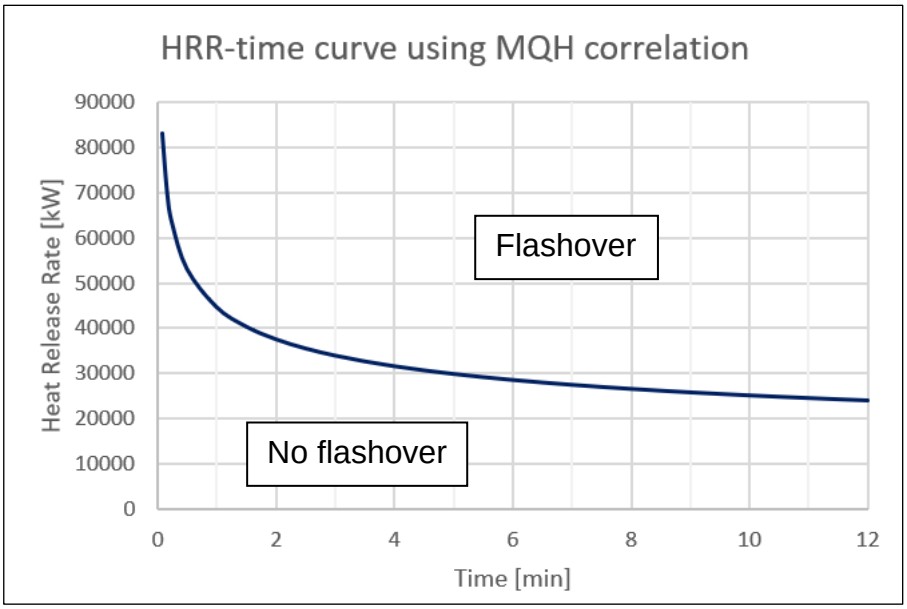


Figure 4.2: MQH derived heat release rate-time curve indicating flashover and no flashover conditions

4.1.2. Travelling Fire Methodology (TFM)

A preliminary spreadsheet has been developed which implements the TFM by Stern-Gottfried and Rein (2012). This preliminary sheet has been used to plot sample temperature-time curves which as shown in Figure 4.3 depict the same features of far-field heating, near-field and post far-field heating noted in the TFM model.

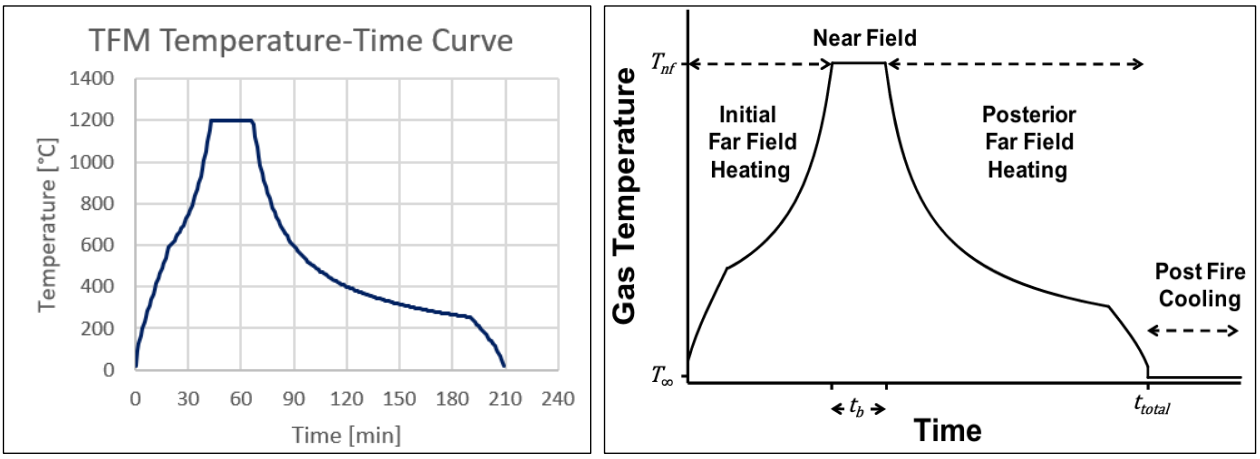


Figure 4.3: Preliminary example TFM temperature-time curve using the developed spreadsheet (left) showing the features of a travelling fire as noted by (Stern-Gottfried & Rein, 2012b) (right)

The developed spreadsheet depicting the noted features of the TFM in addition to the overall shape of the curve indicates that the TFM has been accurately implemented for a single point. The next steps in the implementation of the TFM shall be further refine the process of calculating and graphing the three points proposed such that holistic range of temperature-time curves is investigated.

5. Risk Management, Project Timeline & Conclusions

5.1. Risk Management

The preliminary work and planning for the remainder of the project has considered the potential risks that may affect the execution of the project. Overall, the risks to completion of the project are considered low with the main identified risks able to be categorised into two categories:

- Data analysis risks
- Project management and organisational risks

5.1.1. Data Analysis Risks

One risk to the project identified is the verification of the limitations against experimental evidence. A comprehensive literature review focused on results from experiments has not yet been undertaken and thus it is not known if comparing temperatures measured from experiments against the outputs of the TFM is a reasonable way to verify if the model is outside its appliable limits.

Furthermore, the ability to independently parameterise individual variables is expected to create challenges in determining strict limits. The current proposal to investigate different parameters independently with other variables held at constant base values has the potential limitation of not accounting for the independency of parameters.

The identified risks are not expected to obstruct the completion of the project, but rather the accuracy and potential limitations of any conclusions reached. These risks are considered commonplace in any research and thus shall be addressed through careful documentation and emphasis of any assumptions and shortcoming of the results.

5.1.2. Project Management & Organisational Risks

The other main risks to the project are considered to be organisational in nature and involve the management of the project.

The large range of input parameters to be investigated requires multiple iterations of different methods to be undertaken and thus requires significant time to develop and also results in significant amounts of data to be analysed. The challenges of the data analysis have been discussed previously, however the amount of data proposed to be investigated also creates a risk with respect to the organisation and management of this data to efficiently analyse.

The management strategies employed to mitigate these risks have involved the planning and development of a project timeline. In event of other unexpected circumstances impacting the ability to manage the proposed work, it has also been identified that individual methods such as the fTFM could be removed from the scope of work to ensure the project is completed on time.

5.2. Project timeline

To assist the project planning and organisation of the project, a timeline of key tasks has been developed. The project shall follow the key methodology steps as outlined in Figure 3.1. The key milestones shall be as follows:

Table 5.1: Key deliverables and dates

Task	Milestone	Date
Literature review	Complete literature review with experimental observations and	08/09/24
Spreadsheet development	Working models of each methodology	14/07/24
Input parameter variation	Data collection pertaining to variation of each parameter.	08/09/24
Results Analysis	Limitations from results and correlations with literature identified.	29/09/24

Each of the tasks/milestones listed above shall associated documentation which shall be undertaken during the completion of each task.

The initial timeline and duration of each task is shown Figure 5.1, however this project plan shall be reviewed and updated as required.

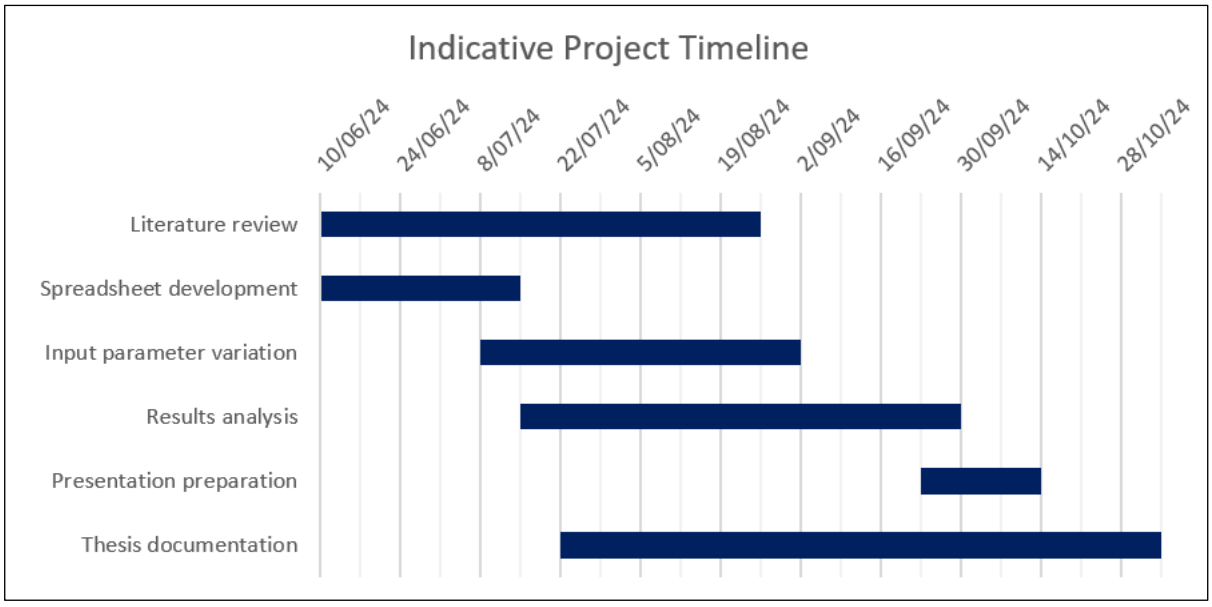


Figure 5.1: Indicative timeline for project tasks

A more detailed project plan is also provided in Appendix A. Detailed Project Timeline.

5.3. Preliminary Conclusions

The current work has provided a literature review into compartment fires and travelling fires which form the foundations of this project. The literature review has investigated the different travelling fire models developed and identified the different input parameters required. The MQH correlation has also been investigated and is proposed as a method to conduct a preliminary check of any proposed parameters.

Early implementations of the MQH correlations and TFM have shown reasonable similarities with the documented results in literature sources thus indicating early spreadsheets are accurately configured.

The identified risks have been assessed and been deemed to pose an overall low risk to the project completion, nonetheless mitigation strategies have been developed to address these risks. As part of this risk management and project planning, a timeline of tasks pertaining to the project to ensure its delivery has also been developed.

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Appendix A. Detailed Project Timeline

Table A.1: Project Timeline

Tasks	10/06	17/06	24/06	01/07	08/07	15/07	22/07	29/07	05/08	12/08	19/08	26/08	02/09	09/09	16/09	23/09	30/09	07/10	14/ 10	21/ 10	28/ 10	04/ 11
Literature Review																						
Spreadsheet development																						
MQH																						
TFM																						
iTFM																						
fTFM																						
Verification																						
Input parameter variation																						
Compartment Floor Area																						
Length-to-width aspect ratio																						
Compartment Height																						
Ventilation Opening																						
Fire Size																						
Fuel Load Density																						
HRRPUA																						
Results Analysis																						
Presentation preparation																						
Thesis documentation																						
Milestone/Deliverable	MQH Model	TFM Model	iTFM Model	fTFM Model						Geometry parameter complete			Fire Parameter complete					Presentation slides Draft Thesis	Final Presentation			Thesis Submission